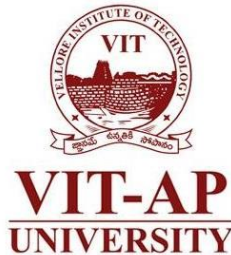


MAT1001 Calculus for Engineers – An Outline



Department of Mathematics
School of Advanced Sciences
Vellore Institute of Technology – Andhra Pradesh

- Course Code : MAT1001
- Course Title : Calculus for Engineers
- Version : 2.0
- Credits : 4 [TPC: 3 2 4]
- Prerequisites : Nil
- Anti-requisites : MAT1008

- Slots Allotted : Theory: G1, TGG1, Lab: Will be assigned separately
- 3 Theory Hours + 2 Lab Hours / Week
- Theory Hours : **Tuesday 12:00 – 12:50 pm**
Friday 09:00-09:50 am
Saturday 10:00 – 10:50 am

At the end of this course, which emphasizes mathematical modeling, analysis and use of MATLAB, the students are expected to learn

- How to evaluate multiple integrals in Cartesian, Cylindrical and Spherical geometries.
- Chain rules for partial differentiation, methods for finding maxima and minima and technique of linearization.
- The powerful language of Vector calculus with physical understanding to deal with subjects such as Fluid Dynamics and Electromagnetic fields.
- How to represent time functions in terms of Eigen functions
- Mathematical modeling, analysis & interpretation of the results, and graphical visualization through MATLAB

Total Number of Modules : 5

- Module 1 : Single Variable Calculus
- Module 2 : Multivariable Differential Calculus
- Module 3 : Multivariable Integral Calculus
- Module 4 : Vector Calculus
- Module 5 : Fourier Series

Total Number of Modules : 5

- Module 1 : Single Variable Calculus

(Review of Limit, continuity of functions of single variable)

Differentiation- Extrema on an Interval Rolle's Theorem and the Mean value theorem-Increasing and decreasing functions-First derivative test-Second derivative test-Maxima and Minima-Concavity. Integration-Average function value - Area between curves - Volumes of solids of revolution, Convergence of Sequences and Series.

Total Number of Modules : 5

- **Module 2 : Multivariable Differential Calculus**

Functions of two or more variables, Level curves; limits and continuity-partial derivatives-Chain Rules–Total differential; Jacobian, Taylor's expansion for two variables-Linear Approximation and Increment estimation; Maxima, minima and saddle points, Constrained maxima and minima-Lagrange's multiplier method, Differentiation of integrals containing a parameter-Leibnitz's rule.

Total Number of Modules : 5

- **Module 3 : Multivariable Integral Calculus**

Evaluation of double integrals—change of order of integration— change of variables between Cartesian and polar co-ordinates- evaluation of triple integrals-change of variables between Cartesian and cylindrical and spherical polar co-ordinates; beta and gamma functions—interrelation-evaluation of integrals using gamma and beta functions; error function-properties.

Total Number of Modules : 5

- **Module 4 : Vector Calculus**

Scalar and vector point functions - Vector Differentiation- expressions for velocity and acceleration in cylindrical geometry. Gradient–physical interpretation-total derivative–directional derivative; Divergence and Curl - physical interpretations-Statement of vector identities - scalar and vector potentials-line, surface, and volume integrals- Statement of Green's, Stoke's and Gauss divergence theorems - verification and evaluation of vector integrals using them.

Total Number of Modules : 5

- Module 5 : Fourier Series

Fourier series -Euler's formulae- Dirichlet's conditions - half range series – RMS value – Parseval's identity.

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- Module-1: Distance, Velocity, Acceleration of moving object, Area under a curve, Work-Energy principle, Arc length and area of Surface of revolution, Volume of surface of revolution.
- Module-2: Rate of change-Electrical circuits, Ideal gas and Fluid flow, Maxima and minima-Optimization.
- Module-3: Mass of the lamina, Total charge, Area bounded by plane curves, Average value, Moment of inertia, Center of mass/gravity, Centroid.
- Module-4: Maximum variation of the surface, Flux, Circulation, Work done by the force, Conservative fields.
- Module-5: Time signal and Voltage pulse in terms of Fourier series, Power of the signal.

Understanding of the concepts through mathematics lab - 10 experiments

- 1 Introduction to MATLAB through matrices
2. Plotting and visualizing functions, rates of change of functions / tangent line
3. Understanding integration as area under the curve
4. Evaluating extremum of a single variable function
5. Plotting of surfaces and visualizing tangent planes
6. Evaluating maxima and minima of function of several variables
7. Applying Lagrange Multiplier Method
8. Evaluating volume under surfaces
9. Evaluating triple integrals
10. Evaluating integrals using cylindrical and spherical coordinates and their visualization
11. Evaluating gradient, curl and divergence
12. Evaluating line integrals
13. Applying Green's, Stoke's and Gauss divergence
14. Fourier series of time function – Visualization

Prescribed Textbook

- George B. Thomas, D. Weir and J. Hass (2013), Thomas' Calculus, 13^h edition, Pearson.

References

- Gilbert Strang (2010), Calculus, 2nd Edition, Wellesley-Cambridge Press.
- Dennis G Zill and Warren S Wright (2011), Multivariable Calculus, 4th Edition, Jones and Bartlett Publishers.
- Erwin Kreyszig (2017), Advanced Engineering Mathematics, 10th Edition, John Wiley.
- Glyn James (2016), Advanced Modern Engineering Mathematics, 4th Edition, Pearson.
- B. S. Grewal (2017), Higher Engineering Mathematics by, 44th Edition, Khanna Publishers.
- Rudra Pratap(2016), Getting started with MATLAB: A quick introduction for scientists and Engineers, 7th Edition, Oxford University Press.

Course Outcomes	Course Outcome Statement	PO's
CO1	To use the concepts of limit, differentiation and integration of functions of single variable for extreme values, average value of these function and to find area between curves.	PO1, PO2, PO3, PO5
CO2	To use the partial differentiation for finding rate of change, maxima & minima, linearization	PO1, PO2, PO3, PO5,
CO3	To evaluate multiple integrals in Cartesian, Cylindrical and Spherical geometries, and apply it to engineering problems.	PO1, PO2, PO3, PO5,
CO4	To apply vector calculus in various subjects like Fluid Dynamics and Electromagnetic fields.	PO1, PO2, PO3, PO5, PO6, PO12
CO5	To apply Fourier series to mechanical systems and electric circuits.	PO1, PO2, PO3
CO6	To construct mathematical modeling, analyze graphical visualization and interpret results through MATLAB.	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PO12

60% : CAT 1 + CAT 2 + FAT (20% each)

CAT1 & CAT2 : 50 Marks, 1 hour 30 minutes

FAT : 60 Marks, 2 hours

40% : Internal Evaluation

Quizzes/ Surprise tests/ Viva/ Presentation/ Notes :15%

Lab Component – Lab Records/ Class Performance/ Quizzes : 25%

Grading Policy: Relative Grading

Pass Criteria:

40% in CAT + FAT (64/160, 160 = 50+50+60)

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At least E grade as per grading criteria

Academic Calendar – Fall Sem. 2023 – 24 (Freshers)

Academic Calendar FALL 2023-24 Semester - Freshers		
Description	Period	Remarks
Freshers' Orientation Programme	08 th September - 12 th September 2023	-----
Commencement of FALL 2023-24 Semester for Freshers	13 th September 2023 (Wednesday)	First Class Day
Vinayaka Chaturthi	18 th September 2023 (Monday)	No Class Day
Milad-un-Nabi	28 th September 2023 (Thursday)	No Class Day
Mahatma Gandhi Jayanti	02 nd October 2022 (Monday)	No Class Day
CAT-1	14th October - 21st October 2023	Exam Days
Vijayadasami/Dussehra	24 th October 2023 (Tuesday)	No Class Day
Deepavali	12 th November 2023 (Sunday)	No Class Day
CAT-2	30th November - 07th December 2023	Exam Days
Christmas	25 th December 2023 (Monday)	No Class Day
Winter vacation for students	26 th December 2023 - 02 nd January 2024	No Class Day
New Year	01 st January 2024 (Monday)	No Class Day
Bhogi	14 th January 2024 (Sunday)	No Class Day
Makara Sankranthi/ Pongal	15 th January 2024 (Monday)	No Class Day
Republic Day	26 th January 2024 (Friday)	No Class Day
FAT	27th January - 03rd February 2024	Exam Days
End of FALL 2023-24 Semester	03 rd February 2024 (Saturday)	-----
Tentative Commencement of WIN 2023-24 Semester	06 th February 2024 (Tuesday)	-----

Name: Dr. Aswathy R K

Contact details:

Email : aswathy.rk@vitap.ac.in

Mobile : 9446952391

Room Number: 312, CB

Department: Mathematics

School: School of Advanced Sciences

The logo of VIT-AP University is a circular emblem. It features a central image of a temple gopuram (tower) with a flag on top, set against a background of a body of water and a city skyline. The text "VIT" is prominently displayed in the center of the emblem.

Thank You...

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